Physical Strip Attack for Object Detection in Optical Remote Sensing

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***Abstract*— A growing trend in the field of adversarial attacks is evolving from the digital domain to the more challenging physical domain. The previous works mainly employ printable adversarial patches with special textures in real-world physical attacks. However, due to lighting conditions and atmospheric scattering, the texture-based patches are prone to distortion in the long-range situation than in the close-range case, resulting in poor physical attack performance in remote sensing scenarios. Therefore, this article proposes a new physical attack method using single-color strip-based patches to hide the objects from being detected correctly in optical aerial detection. Specifically, we design a differentiable representation and an optimization method to optimize the position, thickness, and color of the adversarial strips. Compared with the traditional complex texture-based patch, the proposed strip-based patch is more robust when mapping from the digital domain to the physical domain. Extensive experiments are conducted on multiple datasets and real-world scenarios to evaluate the attack performance of various attack methods. The results show that the proposed strip-based adversarial patch has better attack performance against white-box, black-box, and even defense detectors. Furthermore, we can improve the physical attack success rate (ASR) in remote sensing scenarios by about 70% compared with previous texture-based methods.**

***Index Terms*— Physical adversarial attack, remote sensing detection, strip-based adversarial patch.**

I. Introduction

THE deep neural network (DNN)-based object detection has shown remarkable performance in accuracy and efficiency with the rapid development of deep learning [1]. However, many studies [2], [3] demonstrate that DNNs are vulnerable to elaborate on adversarial examples, i.e., adding malicious perturbations or patches into clean examples can easily fool object detect models. According to extensive investigations, the earlier works mainly focus on digital attacks or physical attacks in near-distance scenarios, such as face recognition [4], [5] and autonomous driving [6], [7]. As the objects in remote sensing are much smaller in long-distance scenarios, the adversarial attack against aerial detection is more challenging, especially in the physical world.

Manuscript received 17 April 2024; revised 1 July 2024; accepted 23 July 2024. Date of publication 29 July 2024; date of current version 19 September 2024. This work was supported in part by the Key Research and Development Project in Shaanxi Province under Grant 2023-ZDLNY-65 and in part by the Central Guidance on Local Science and Technology Development Fund under Grant 2022ZY1-CGZY-01HZ01. *(Corresponding author: Xuchong Zhang.)*

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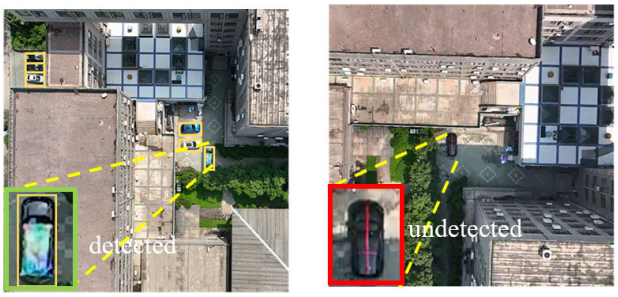
Digital Object Identifier 10.1109/TGRS.2024.3434430

The adversarial example mainly contains two types: human-imperceptible perturbations [8] and image patches [3]. The former usually manipulates each pixel in the whole image, which is applicable to digital attacks in web applications. The latter is printable and can be attached to objects; thus, it is the primary manner of the adversarial example that was applied in real-world scenarios. Recently, a growing trend is that a large amount of research [9], [10] is targeting physical adversarial attacks. Typically, they are implemented by optimizing adversarial patches with special and complex textures, as shown in Fig. 1(a). Although these methods can deceive detection models, the adversarial attacks are only effective at close range (e.g., a few meters or tens of meters). For the long-range physical attack even at hundreds of meters in remote sensing, there are relatively few studies, and they are directly derived from the physical attack approaches of general scenarios. Because the image quality and color fidelity of texture-based adversarial patches are susceptible to lighting conditions, atmospheric scattering, and refraction in long-range situations, these texture-based adversarial examples for remote scenarios are distorted, and the physical attack performance will drop significantly. As presented in [11], the texture-based patches for attacking YOLO in remote sensing achieve a 50% attack success rate (ASR) at 50 m but totally lose the attack ability when the distance exceeds 100 m. Since the long distance between the camera and object is a fundamental characteristic in remote sensing scenes, the low-quality and distorted image is inevitable. Therefore, even if we design a patch with remarkable attack performance, it will lose its attack ability in the physics implementations due to the distortion. In this context, developing an efficient physical adversarial attack method suitable for low-quality and distorted images in remote sensing presents a significant challenge.

For DNN-based object detection models, texture information indeed provides important features, and this may be the main reason why current studies mislead detectors by changing salient features using texture-based adversarial patches. Nonetheless, in addition to texture details, relying on contour and edge information to locate the object is also very important to detect objects [12]. Inspired by this mechanism, we propose a new physical adversarial attack method using strip-based patches to hide the objects from being detected in remote sensing, as shown in Fig. 1(b). The key idea is to mislead the prediction of detectors' location heads by attaching the optimized single-color strips on objects. Once the object location is incorrectly predicted, the performance

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(a) (b)

Fig. 1. Traditional texture-based adversarial patch versus the proposed strip-based adversarial patch. The latter has much higher adversarial attack performance than the former in the physical domain. (a) Texture-based patch and (b) strip-based patch.

of the object detector will decrease significantly. Concretely, we design a differentiable representation approach and an optimization method to update the position, thickness, and color of the adversarial strips. Compared to the complex textures, the proposed single-color strips method minimizes the distortion in the physical image process, thereby ensuring high attack performance in the physical domain. We carry out extensive experiments on the DOTA/NWPU VHR-10 datasets and real-world scenarios to evaluate the effectiveness of various attack methods. The results show that the adversarial strips outperform the previous texture-based adversarial patches against multiple white-box, black-box, and defense detectors. More importantly, the physical attack performance of the proposed method keeps high ASRs (∼70%) even in the remote sensing scenarios at a height of 300 m, whereas the texture-based adversarial patches have little attack effect.

The contributions of our work are summarized as follows.

1) We propose a new physical attack method for remote sensing scenarios, which optimizes single-color adversarial strips rather than the traditional texture-based adversarial patches.
2) In order to obtain the optimal adversarial strips, a differentiable representation and an optimization method are elaborated to update the critical information including the position, thickness, and color of the strips.
3) The experiments on multiple datasets and the real world show that the proposed method largely improves the physical attack performance in remote sensing compared with previous texture-based adversarial patches. To the best of our knowledge, we are the first to achieve high physical ASR in remote sensing.

The remainder of this article is organized as follows. Section II briefly reviews the related work of physical adversarial attacks. Then, we introduce the details of the proposed framework for generating strip-based adversarial patches against aerial detection tasks in Section III. We evaluate the proposed attack method and demonstrate its effectiveness in Section IV. Finally, further discussions and conclusions are summarized in Section V.

II. Related Work

In this section, we first review the works about remote sensing image processing and analysis, especially object detection. Then, we introduce the physical attacks for object detection in terms of near-distance scenarios and long-distance scenarios.

A. Remote Sensing Image Processing and Analysis

With the rapid advancement of deep learning, significant progress has been made in the processing and analysis of remote sensing images. For example, Hong et al. [13] create a universal foundation model, named SpectralGPT, to handle spectral remote sensing images using a novel 3-D generative pretrained transformer and achieve significant performance improvements in four downstream tasks. As one of the fundamental tasks in remote sensing image analysis, object detection has shown remarkable performance with DNNs. Based on the typical object detection structure in computer vision, researchers add an extra branch to predict the rotation angle of objects in remote sensing scenes and create several networks, such as YOLOv5 OBB [14], RetinaNet OBB [15], and Faster RCNN OBB [16]. In order to break the computational bottleneck of advanced oriented object detection, Xie et al. [17] propose an effective and simple oriented object detection framework, termed Oriented RCNN, promising both accuracy and efficiency. Han et al. [18] propose S2Anet to solve the misalignment between anchor boxes and axis-aligned convolutional features. Ding et al. [19] propose an RoI Transformer to address the common misalignment between the final object classification confidence and localization accuracy. Although these works achieve high performance with the DNNs, many studies [2], [3] demonstrate that DNNs are vulnerable to elaborate on adversarial examples. In this article, we study the vulnerability of object detection in remote sensing scenario and attempt to propose a method for physical attacks.

B. Physical Adversarial Attack in Near-Distance Scenarios

Most of the previous physical adversarial attacks focus on near-distance scenarios, such as autonomous driving and person detection. Chow et al. [3] proposed a general attack method for object detection, which can hide objects in all classes. They optimized texture-based patches by reducing the objectness score of detected objects and attached the trained patches on objects to deceive detectors. Besides the general attack methods, many researchers proposed methods to attack specific objects. Chen et al. [2] and Zhao et al. [20] attached the adversarial patches or modified the texture of the stop sign to mislead the detectors into ignoring the stop sign or misclassifying it as a limited speed sign. Wang et al. [10], [21] used a neural render to integrate adversarial patches onto cars and deceived detectors effectively. Hu et al. [9] and Xu et al. [22] utilized specific patterns to create adversarial clothing, achieving an invisibility effect against person detectors.

C. Adversarial Attack in Long-Distance Scenarios

Recently, some recent works aim to design elaborate adversarial examples for optical aerial detection scenes in digital and physical domains. Sun et al. [23] proposed a threatening patch attack with the form of human-imperceptible perturbations to achieve remarkable results in digital benchmarks of optical

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remote sensing images. Tang et al. [24] conducted adversarial attacks against optical aerial detectors by leveraging novel natural weather-style perturbations. The results evaluated on the DOTA dataset [25] show the effectiveness of black-box adversarial attacks, but it is challenging to add weather noise (such as fog or sun flare) in real-world scenarios.

Similar to near-distance scenarios, all existing physical attack methods optimized texture-based patches to fool optical remote sensing detection models by attaching patches on or near the object. Zhang et al. [11] proposed a texture-based patch to attack YOLO by minimizing objectness scores of target objects and maximizing the detection loss of detectors, achieving the ASRs of 50% and 9% at 50 and 100 m, respectively. Du et al. [26] devised a physical attack method by attaching the texture-based patches near target objects to attack object detectors in aerial scenes. Tang et al. [27] designed a loss function using the intermediate outputs of the models rather than the model's final outputs to optimize texture-based adversarial patches. Lian et al. [28] proposed a contextual background attack framework to fool aerial detectors in the physical world by changing the texture of the critical contextual background area rather than smudging the victim objects.

Although these methods have achieved a certain ASR, real-world experiments are mainly conducted on toy models or a distance of no more than 80 m between objects and cameras. Once the distance increases to a few hundred meters, it is hard to guarantee the attack effectiveness since the color fidelity of texture-based adversarial patches is susceptible to lighting conditions, atmospheric scattering, and refraction in the physical domain. This motivates us to investigate a new physical attack method that is more practical and feasible in long-distance remote sensing scenarios.

III. Method

In this article, a new adversarial attack method employing single-color strips is proposed to attack object detection in remote sensing scenarios. Specifically, we design a differentiable representation and an optimization method to find the optimal position, thickness, and color for the adversarial strips.

A. Problem Formulation

Given a clean image $\mathbf{I}$, a well-trained object detector $\mathcal{G}$ can correctly detect the objects contained in the image, i.e., $\mathcal{G}(\mathbf{I}) = \mathcal{O}$, where $\mathcal{O}$ is the set of detected objects $\{\mathbf{o}_1, \mathbf{o}_2, \mathbf{o}_3, \ldots, \mathbf{o}_n\}$. Each detected object $\mathbf{o}_i$ can be represented by a tuple $\{x, y, w, h, \text{cls}, \text{confi}\}$, where x and y represent the coordinates of the bounding box center, and w and h represent the width and height of the bounding box, while cls and confi represent the predicted class label and the confidence score of the prediction. The proposed attack method attempts to find the optimal adversarial strips $\mathbf{P} = \{\mathbf{P}_1, \mathbf{P}_2, \ldots, \mathbf{P}_m\}$, which are attached on the object $\mathbf{o}_k$ to construct the adversarial example $\mathbf{I}^{\text{adv}}$. Then, $\mathbf{I}^{\text{adv}}$ can be used to deceive the object detector into missing the detection of the object $\mathbf{o}_k$. We note that m is a hyperparameter indicating the maximum number of strips to be optimized. In the following contents, we take the optimization of one strip, i.e., $m = 1$ as the example to present our method in detail. For the case of multiple strips, i.e., $m > 1$, it only needs to optimize multiple strips in parallel. The optimization of one strip $\mathbf{P}$ can be formulated as

$$\mathbf{P} = \arg\min \mathcal{L}\left(\mathcal{G}, \mathbf{I}^{\text{adv}}, \mathbf{o}_k\right), \quad \text{s.t. } \mathcal{G}\left(\mathbf{I}^{\text{adv}}\right) = \mathcal{O} - \{\mathbf{o}_k\}. \tag{1}$$

The final adversarial example $\mathbf{I}_{\text{adv}}$ can be denoted as

$$\mathbf{I}^{\text{adv}} = \gamma \cdot \mathbf{M} + (\mathbf{1} - \mathbf{M}) \odot \mathbf{I} \tag{2}$$

where γ represents the single color of the adversarial strip, $\mathbf{M} \in \{0, 1\}^{c \times h \times w}$ is a binary mask with the same dimension of $\mathbf{I}$, and $\gamma \cdot \mathbf{M}$ represents the adversarial strip $\mathbf{P}$. The value 1 within $\mathbf{M}$ denotes the pixel at the corresponding position belonging to the adversarial strip in $\mathbf{I}^{\text{adv}}$. It is obvious that $\mathbf{M}$ contains the position and thickness information of the strip. Then, we aim to obtain the optimal adversarial strip by optimizing γ and $\mathbf{M}$. It should be emphasized that in previous works, $\mathbf{M}$ is fixed over the entire attack progress because these methods only optimize the RGB value of each pixel of the texture-based patch, and its position is preset before conducting attack [29], [30].

B. Differentiable Representation of Strip

As shown in (2), the adversarial strip is determined by the color γ and mask matrix $\mathbf{M}$. γ indicates that the strip has uniform RGB values varying from 0 to 255. However, the $\mathbf{M}$ is a nondifferentiable binary matrix; thus, it cannot be updated by the gradient-based optimization process. Therefore, it is necessary to design a differentiable representation to approximate $\mathbf{M}$, so as to realize the backpropagation.

The left part of Fig. 2 illustrates the representation of an adversarial strip. It is composed of pixels whose distance from the center line (red line) is less than the strip's half thickness; thus, the mask matrix $\mathbf{M}$ can be calculated as

$$\mathbf{M}_{x,y} = \begin{cases} 1, & d_{x,y} < s \\ 0, & d_{x,y} > s \end{cases} \tag{3}$$

where $d_{x,y}$ represents the distance between the pixel (x, y) and center line, while s denotes half the thickness of the strip. To approximate $\mathbf{M}$ in a differentiable manner, we first compute the distance of each pixel from the center line and use a differentiable function to map the distance to 0 or 1.

For the optimization of only one adversarial strip, the center of the object is selected as the initial center point (x_0, y_0) of the strip. Thus, the center line of the strip is formulated as

$$(\mathbf{x} - x_0) + k(\mathbf{y} - y_0) = 0 \tag{4}$$

where k is the slope parameter. We can compute the distance of each pixel as follows:

$$d_{x,y} = \frac{|x - x_0 + k(y - y_0)|}{\sqrt{1 + k^2}}. \tag{5}$$

Next, a differentiable function is designed to map the distance as

$$\psi\left(\frac{d}{s}\right) = \frac{\tanh\left(\rho\left(\frac{d}{s} - 1\right)\right) + 1}{2} \tag{6}$$

where ρ is set to -100, and the right part of Fig. 2 illustrates the mapping results. We can see that if (d/s) is less than 1,

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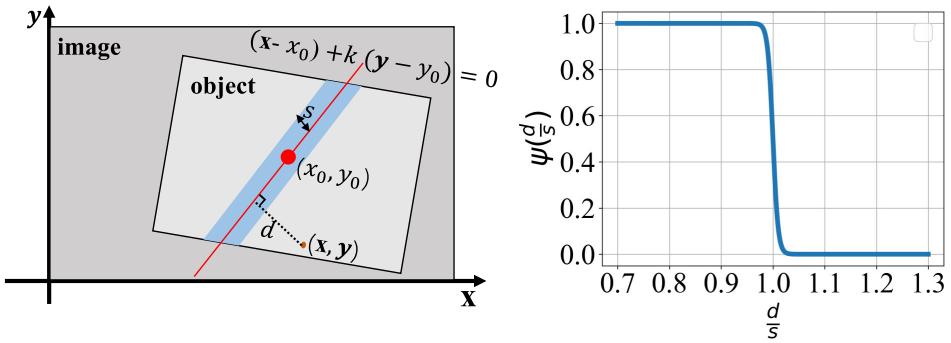


Fig. 2. Differentiable representation of adversarial strip. It can be represented by a center point (x_0, y_0), slope parameter k, thickness s, and color γ in a differentiable manner.

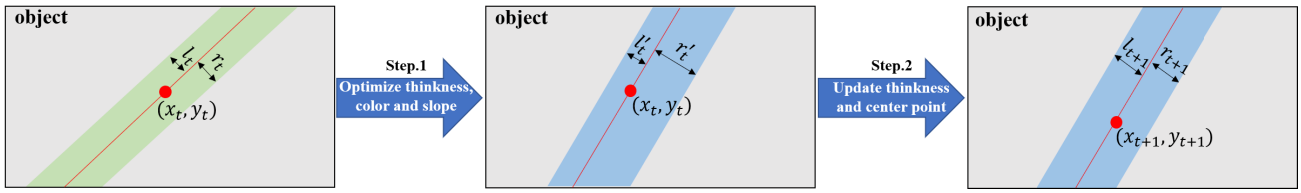


Fig. 3. Optimization process of the adversarial strip. Each iteration mainly contains two steps: the first step optimizes the color parameter γ, slope parameter k, and thickness parameter (l, r), and the second step updates the center point (x, y) and thickness (l, r) of the adversarial strip.

the function approximately outputs 1, while if (d/s) is greater than 1, the function approximately outputs 0. Finally, each value in the approximate matrix can be calculated as

$$\hat{\mathbf{M}}_{x,y} = \psi\left(\frac{d(x, y)}{s}\right). \tag{7}$$

Because the above function $\psi(\ldots)$ is continuous and differentiable, it can be used to propagate gradients during the optimization process. Accordingly, the original binary mask $\mathbf{M}$ is approximately replaced by the combination of center point (x_0, y_0), slope parameter k, and thickness s in a differentiable representation, where (x_0, y_0) and k determine the position of the adversarial strip in adversarial examples.

C. Optimization Process of Strip

Based on the above representation, the optimization of the adversarial strip is equivalent to the optimization of the position variables (x_0, y_0) and k, color γ, and thickness s. In these parameters, the values of γ, k, and s are all continuously varying within a certain range, which can be directly optimized by the gradient-based optimization process. Nevertheless, the center point (x_0, y_0) presented in Section III-B is a preset value, which means that this adversarial strip only optimizes color, thickness, and slope in a fixed point. To further improve the attack performance of the proposed method, we design an update method to iteratively alter the position of the center point of the adversarial strip.

Fig. 3 illustrates the optimization procedure from iteration t to $t+1$. First, we divide the strip thickness into left and right parts split by the center line, which are denoted as l and r, respectively. The center point (x_t, y_t) and thickness parameters (l_t, r_t) are the updated results of iteration t. At the beginning of iteration $t+1$, l_t and r_t are equal. During the optimization process, l_t and r_t may be different, and the center point should move toward the direction of the larger value. Specifically, as shown in Fig. 3, assuming that r_t becomes larger (i.e., r'_t) and l_t becomes smaller (i.e., l'_t) after the optimization, we update the center point (x_t, y_t) to the lower right direction along the tangent of the center line until l'_t is equal to r'_t again. The updating procedure of center point (x_t, y_t) is formulated as

$$\begin{cases} x_{t+1} = x_t - \text{sign}(k_{t+1})\dfrac{l'_t - r'_t}{2\sqrt{1+k_{t+1}^2}} \\ y_{t+1} = y_t - \dfrac{k_{t+1}(l'_t - r'_t)}{2\sqrt{1+k_{t+1}^2}}. \end{cases} \tag{8}$$

After moving the center point, the thickness should also be updated as

$$\begin{cases} l_{t+1} = \dfrac{l'_t + r'_t}{2} \\ r_{t+1} = \dfrac{l'_t + r'_t}{2}. \end{cases} \tag{9}$$

In summary, the optimization process in one iteration mainly contains two steps. The first step is to optimize the color parameter γ, slope parameter k, and thickness parameter (l, r) by minimizing the confidence $\mathbf{o}_k^{\text{confi}}$ of the object $\mathbf{o}_k$, i.e., the loss function is presented as

$$\mathcal{L}(\mathcal{G}, \mathbf{I}^{\text{adv}}, \mathbf{o}_k) = \mathbf{o}_k^{\text{confi}}. \tag{10}$$

Then, the second step is to update the center point and thickness of the adversarial strip, as shown in (8) and (9).

After each iteration, we check the values of these parameters and clip them to a reasonable range

$$\begin{aligned} \gamma_{t+1} &= \text{Clip}(\gamma_{t+1}, 0, 255) \\ l_{t+1} &= \text{Clip}(l_{t+1}, 1, z) \\ r_{t+1} &= \text{Clip}(r_{t+1}, 1, z) \end{aligned} \tag{11}$$

where z represents the max thickness of l and r. The concrete optimization process of the proposed attack method is shown in Algorithm 1.

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Algorithm 1 Optimization Algorithm of the Adversarial Strip

Input: source image $\mathbf{I}$, object detector $\mathcal{G}$, Object $\mathbf{o}_k$
Output: adversarial example $\mathbf{I}^{adv}$

Init parameters center point (x_0, y_0), color γ_0, thickness l_0 r_0, slope k_0
$\mathbf{I}^{adv} \leftarrow \mathbf{I}$
for t $\leftarrow 0$ **to** max_{iter} **do**
  Calculate the distance between each pixel and the central line $d = \frac{|x-x_t+k_t(y-y_t)|}{\sqrt{1+k_t^2}}$
  Calculate the approximate mask matrix of pixels on the left side $\hat{\mathbf{M}}(x, y) = \frac{tanh(\rho(\frac{d}{l_t}-1))+1}{2}$
  Calculate the approximate mask matrix of pixels on the right side $\hat{\mathbf{M}}(x, y) = \frac{tanh(\rho(\frac{d}{r_t}-1))+1}{2}$
  Construct adversarial example $\mathbf{I}_t^{adv} = \gamma \cdot \hat{\mathbf{M}} + (\boldsymbol{I} - \hat{\mathbf{M}}) \odot \mathbf{I}$
  Get confidence of $\mathbf{o}_k$, $\mathbf{o}_k^{confi} = \mathcal{G}(\mathbf{I}^{adv})$
  Optimize γ_t, k_t, l_t and r_t by minimizing $\mathcal{L}(\mathcal{G}, \mathbf{I}^{adv}, \mathbf{o}_k) = \mathbf{o}_k^{confi}$ and get γ_{t+1}, k_{t+1}, l'_t, r'_t
  Update center point (x_t, y_t): $x_{t+1} = x_t - sign(k_{t+1})\frac{l'_t - r'_t}{2\sqrt{1+k_{t+1}^2}}$, $y_{t+1} = y_t - \frac{k_{t+1}(l'_t - r'_t)}{2\sqrt{1+k_{t+1}^2}}$
  Update l_t, r_t: $l_{t+1} = \frac{l'_t + r'_t}{2}$, $r_{t+1} = l_{t+1}$
  Project the γ_{t+1}, l_{t+1}, r_{t+1}: $\gamma_{t+1} = Clip(\gamma_{t+1}, 0, 255)$, $l_{t+1} = Clip(l_{t+1}, 1, z)$, and $r_{t+1} = Clip(r_{t+1}, 1, z)$
end for
return adversarial example $\mathbf{I}^{adv}$

IV. Experiment

In this section, we carry out extensive experiments to evaluate the effectiveness of the proposed attack method against white-box, black-box, and defense detectors in the digital and physical domains. Besides this, we also analyze the impact of strips' different factors and explore the attacking mechanisms of adversarial strips.

A. Experimental Settings

1) Dataset and Metric: The experiments are conducted on DOTAv1 [25] and NWPU VHR-10 [31] datasets. The DOTAv1 dataset is a large-scale dataset designed for object detection in remote sensing. It consists of 2806 aerial images from various sensors and platforms, covering 15 different categories. The NWPU VHR-10 dataset is a remote sensing dataset designed for object detection. It comprises 650 images containing objects across ten different categories.

The ASR is used to evaluate the performance of various attack methods. The ASR is calculated across each dataset, which is defined as the proportion of objects that cannot be detected after the attack, compared to the total number of all attacked objects in all images.

2) Object Detectors: To evaluate the performance of the attack methods on different models, we conduct white-box experiments using YOLOv5 OBB [14] and Oriented RCNN [17] since these are the most representative models in remote sensing for one- and two-stage detectors, respectively. In addition, we perform black-box experiments by applying the adversarial examples generated by YOLOv5 OBB and Oriented RCNN to attack S2Anet [18], RetinaNet OBB [15], Faster RCNN OBB [16], and RoI Trans [19]. It can be noted that the above detectors contain CNN-based and transformer-based backbones, and the suffix "OBB" indicates the modification of the original model for oriented bounding box detection. Besides, we employ two representative defense techniques to the adversarial examples, namely, JPEG squeezing and spatial smoothing [32], and then assess the attack performance of different methods against defense strategies. To ensure that the models are well-trained, we use weights released in official sources [33].

3) Comparison Methods: We compare the proposed method with three previous physical attack methods, including one general attack method TOG [3] and two specific methods designed for remote sensing, i.e., APA [11] and PAAI [27]. These methods optimize the texture of the patch by minimizing the objectness scores or classification scores of the target objects and then attach the patch with specific texture to the objects in order to deceive detectors into missing the prediction of the target objects.

4) Hyperparameter Settings: To ensure the fairness of the comparison, we configure the hyperparameters with the same values for all the methods. Concretely, the confidence threshold for all detectors is set to 0.3. The maximum number of iterations for one adversarial example is set to 50. Moreover, our method involves two other hyperparameters, the maximum thickness parameter z is set to 5, and the maximum number of strips m is set to 4.

B. Attack Performance in Digital World

1) Attack Comparison Against White-Box Detectors: We first conduct the white-box attack in which the architecture and parameters of detectors are accessible to the attackers. The success rates for attacking YOLOv5 OBB and Oriented RCNN are shown in the left part of Table I. It can be observed that the proposed adversarial strips achieve 88.4%/70.1% ASRs and 87.2%/78.9% ASRs for attacking YOLOv5 OBB and Oriented RCNN on the DOTA dataset and the NWPU VHR-10 dataset, respectively. Compared with the texture-based adversarial patches proposed in TOG, APA, and PAAI, our method increases the ASRs by at least 10% in the white-box attack of the digital domain. Fig. 4 illustrates the detection results of the adversarial examples generated by different attack methods. We can see that all the methods successfully hide the objects attached with adversarial patches from being detected by YOLOv5 OBB and Oriented RCNN in the digital domain. The main difference is that the previous methods optimize specific and complex textures, while the proposed method optimizes single-color strips. Furthermore, Fig. 5 shows more subjective illustrations of the proposed strip-based patches, which successfully attack various objects, including vehicles, baseball diamonds, harbors, and soccer diamonds.

2) Attack Comparison Against Black-Box Detectors: The black-box attack performance is evaluated by applying the adversarial examples generated by YOLOv5 OBB and Oriented RCNN to directly attack S2ANet, RetinaNet OBB, Faster RCNN OBB, and RoI Trans detectors. As the results shown in the middle part of Table I, the black-box attack performance is degraded to some extent compared with that of the white-box. In general, the proposed strip-based adversarial patch still keeps ahead in each experiment and maintains ~60% ASRs in the case of black-box attacks.

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TABLE I

DIGITAL ATTACK PERFORMANCE COMPARISONS OF THE PROPOSED METHOD AND PREVIOUS METHODS FOR ATTACKING WHITE-BOX, BLACK-BOX, AND DEFENSE DETECTORS ON DOTA AND NWPU VHR-10 DATASETS. THE ASRS IN BLACK-BOX AND DEFENSE METHODS ARE PRESENTED AS THE FORM OF YOLOV5 OBB/ORIENTED RCNN

Attack	White-box		Black-box				Defense Method	
Method	YOLOv5 OBB	Oriented RCNN	S2ANet	RetinaNet OBB	Faster RCNN OBB	RoI Trans	Squeezing	Smoothing
Results on DOTA Dataset								
TOG	71.1%	34.9%	56.7%/10.9%	57.2%/9.5%	47.9%/14.1%	43.1%/15.6%	32.4%/7.8%	28.3%/10.1%
APA	67.8%	57.0%	55.7%/25.7%	56.2%/28.3%	47.4%/19.6%	42.9%/31.5%	29.1%/24.4%	25.4%/26.5%
PAAI	71.4%	59.5%	56.6%/36.2%	57.1%/34.8%	48.1%/33.9%	43.0%/35.5%	32.2%/35.1%	28.2%/43.5%
Ours	**88.4%**	**70.1%**	**66.2%/55.4%**	**65.9%/60.9%**	**64.6%/57.1%**	**61.8%/46.1%**	**69.8%/53.8%**	**62.2%/66.3%**
Results on NWPU VHR-10 Dataset								
TOG	77.8%	13.1%	55.5%/4.3%	54.1%/5.2%	39.9%/4.3%	23.1%/7.9%	38.1%/4.2%	33.9%/1.5%
APA	72.9%	56.5%	54.9%/25.1%	52.6%/27.3%	38.1%/25.7%	22.4%/12.1%	36.3%/29.7%	31.8%/28.8%
PAAI	77.7%	51.8%	55.4%/32.0%	53.1%/32.8%	40.3%/24.4%	22.9%/17.6%	38.3%/35.4%	33.1%/30.8%
Ours	**87.2%**	**78.9%**	**58.5%/59.9%**	**59.5%/66.7%**	**66.6%/65.4%**	**60.5%/51.1%**	**75.2%/67.2%**	**72.3%/77.6%**

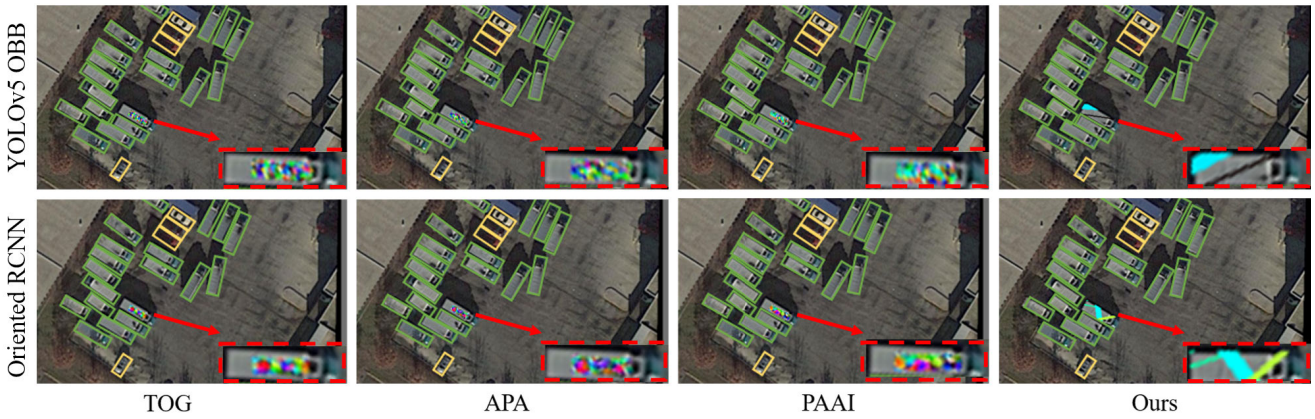


Fig. 4. Detection results of the adversarial examples generated by white-box settings of various attack methods in the digital domain. TOG, APA, and PAAI optimize texture-based adversarial patches, and the proposed method optimizes single-color strip-based adversarial patches.

In contrast, the black-box attack performance of texture-based adversarial patches drops significantly. For instance, TOG, APA, and PAAI only achieve ∼14%, ∼20%, and ∼34% ASRs when applying the adversarial examples generated by Oriented RCNN to attack Faster RCNN OBB on the DOTA dataset. Similarly, the ASRs are ∼23% for these three methods when applying the adversarial examples generated by YOLOv5 OBB to attack the RoI Trans detector on the NWPU VHR-10 dataset. The above experimental results demonstrate that the proposed adversarial strips are not only effective in white-box attack scenarios but also have better transferability than texture-based adversarial patches in black-box attack scenarios.

3) Attack Comparison Against Defense Strategies: We further assess the attack performance of different methods against the commonly used defense strategies in the digital domain. Accordingly, JPEG squeezing and spatial smoothing are utilized to preprocess the obtained adversarial examples separately and then recalculate the ASRs of different attack methods on YOLOv5 OBB and Oriented RCNN.

As shown in the right part of Table I, the ASRs of TOG, APA, and PAAI are almost halved compared with those of white-box attacks. Because the texture features may be corrupted by squeezing or smoothing operations, the attack effects of texture-based adversarial patches are usually susceptible to these defense techniques. In contrast, the proposed method keeps high ASRs even with the defense strategies. On the NWPU VHR-10 dataset, we can still achieve 75.2%/67.2% and 72.3%/77.6% ASRs after applying JPEG squeezing and spatial smoothing, respectively. These results indicate the robustness of the proposed single-color adversarial strips to certain defense strategies.

C. Attack Performance in Physical World

In addition to the above digital attacks, we further conduct experiments in real-world remote sensing scenarios to evaluate the physical attack performance of different methods. Specifically, we select a black SUV and a gray car as the target objects and use a DJI Mini 2 drone to capture video from the ground up to a height of 300 m. In the following experiments, we mainly evaluate the physical attack performance between 120 and 300 m. For the near-distance scenarios, previous works have evaluated the effectiveness of texture-based adversarial patches. Moreover, the detectors employed in our experiments are mainly trained on the DOTA dataset, which is applicable to remote sensing scenarios. For the case of greater than 300 m, the objects in the captured images are too small to be detected.

After obtaining the videos in the real world, we first employ YOLOv5 OBB to optimize the texture-based adversarial patches using TOG, APA, and PAAI and the strip-based

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Fig. 5. Detection results of the strip-based patches, which successfully attack vehicles, baseball diamonds, harbor, and soccer diamonds in the digital domain.



Fig. 6. Detection results of the adversarial examples generated by strip- and texture-based methods for attacking YOLOv5 OBB in the physical domain. The first column visualizes the optimized texture- and strip-based adversarial patches attached to vehicles, while the other three columns show the detection results of different methods at the heights of 120, 200, and 300 m, respectively.

adversarial patches using the proposed method. Then, these adversarial patches are printed and attached to the top of vehicles individually. Finally, we use the drone to capture the real-world scenes again to test the physical attack performance.

We evaluate the physical white-box attack on YOLOv5 OBB and the black-box attack on Oriented RCNN, respectively. The results in Table II are the ASRs based on data captured over a height range from 120 to 300 m. We can see that the proposed adversarial strips method achieves 76.2% and 64.1% ASRs for attacking these two detectors, whereas texture-based adversarial patches have little attack effect. The first column of Fig. 6 visualizes the optimized texture- and strip-based adversarial patches attached to vehicles in real-world scenarios. The other three columns show the detection results of different methods in the height of 120, 200, and 300 m, respectively. It can be observed that the target objects with adversarial strips are successfully hidden from being detected by the YOLOv5 OBB detector, while the target

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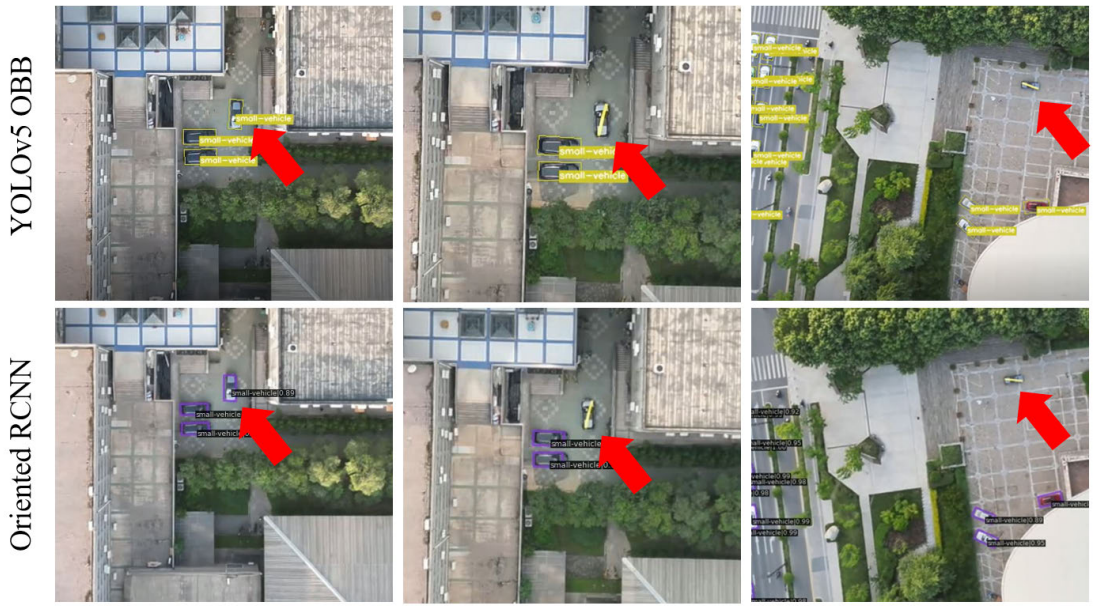


Fig. 7. Detection result of the adversarial examples generated by the proposed method for attacking YOLOv5 OBB and Oriented RCNN in the physical world.

TABLE II

PHYSICAL ATTACK PERFORMANCE COMPARISONS OF THE PROPOSED METHOD AND PREVIOUS METHODS FOR ATTACKING YOLOV5 OBB IN THE WHITE-BOX SETTING AND ORIENTED RCNN IN THE BLACK-BOX SETTING

Attack method	YOLOv5 OBB	Oriented RCNN
TOG	0.3%	0.1%
APA	2.4%	1.5%
PAAI	2.6%	1.3%
Ours	**76.2%**	**64.1%**

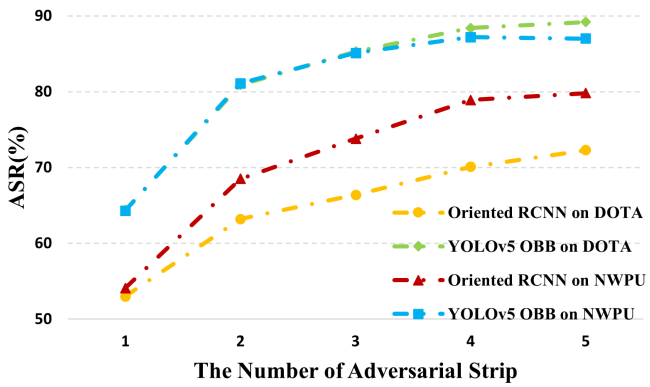


Fig. 8. ASRs of different numbers of adversarial strips.

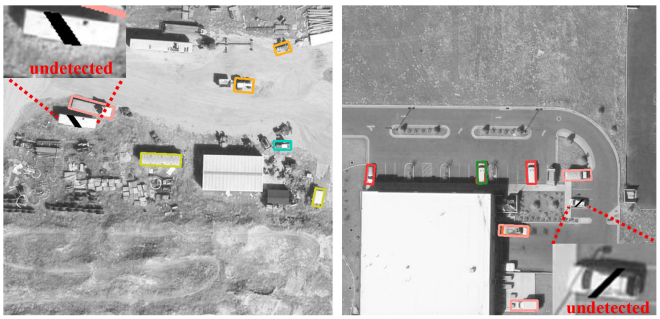


Fig. 9. Detection results of the adversarial examples generated by the proposed methods for attacking YOLOv5 OBB in the digital domain at the infrared scene.

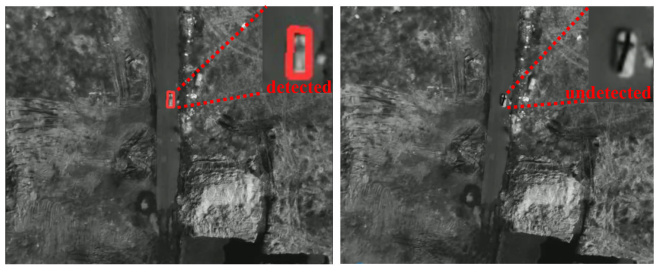
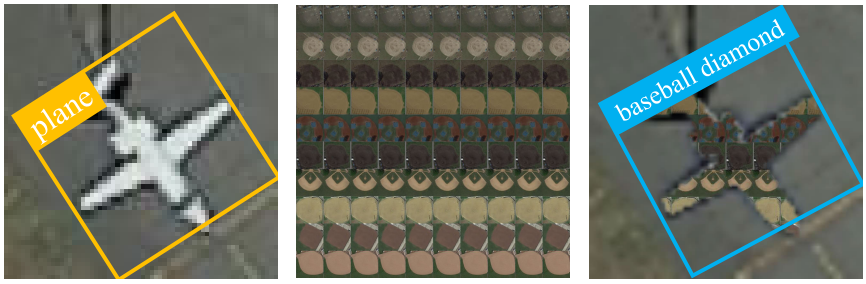


Fig. 10. Detection results of the adversarial examples generated by the proposed methods for attacking YOLOv5 OBB in the physical domain at the infrared scene.

object with a texture-based adversarial patch can still be detected. Moreover, these experiments are conducted in sunny and cloudy conditions, respectively, which confirms that the proposed single-color adversarial strips are more robust than texture-based adversarial patches when taking into account the effects of lighting conditions and atmospheric scattering in the physical world. To the best of our knowledge, we are the first to achieve a high physical ASR in remote sensing scenarios.

We also conduct experiments to verify the attack performance of the same strip to attack different object detectors under different backgrounds. First, we design strips as described above and attach strips to the car, and then, we set the car under different backgrounds. Afterward, we capture the video from the ground up to a height of 300 m and use the YOLOv5 OBB and Oriented RCNN to detect the captured videos. Fig. 7 shows the detection results of the adversarial examples for attacking YOLOv5 OBB and Oriented RCNN under two different backgrounds. The first column shows the detection results of the car without strips; we can see that both two detectors can precisely detect the target car. The second and third columns show the detection results of the cars with strips under two different backgrounds.

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(a) (b) (c)

Fig. 11. Illustration of attacking mechanism analysis for texture-based adversarial patches. (a) Clean image. (b) Texture pattern. (c) Crafted image.

TABLE III
ABLATION STUDY OF THE POSITION, THICKNESS, AND COLOR OF ADVERSARIAL STRIPS

Factors			Model			
Pos.	THK	CLR	YOLOv5 OBB		Oriented RCNN	
			DOTA	NWPU	DOTA	NWPU
-	-	-	35.2%	36.7%	27.8%	29.4%
✓	-	-	59.7%	62.7%	51.3%	59.2%
-	✓	-	64.4%	59.4%	46.9%	53.8%
-	-	✓	60.6%	68.3%	57.7%	63.9%
✓	✓	-	78.8%	70.3%	54.4%	64.5%
✓	-	✓	68.2%	78.4%	62.9%	72.0%
-	✓	✓	74.8%	76.7%	63.2%	68.0%
✓	✓	✓	**88.4%**	**87.2%**	**70.1%**	**78.9%**

After attaching strips to the target car, both two detectors ignore the target car and precisely detect other cars. Therefore, we can draw a conclusion that the strips generated by the proposed method can attack different detectors simultaneously even under different backgrounds.

D. Ablation Study

1) Analysis of the Number of Strip: As introduced in Section III, the optimal adversarial strips $\mathbf{P}$ may contain a set of single-color strips $\{\mathbf{P}_1, \mathbf{P}_2, \ldots, \mathbf{P}_m\}$ after optimization process. Therefore, an analysis of the number of adversarial strips is presented in this section. Fig. 8 shows the white-box ASRs of the proposed method against YOLOv5 OBB and Oriented RCNN in the cases of $m = 1, 2, 3, 4, 5$. As the number of adversarial strips increases, the ASR is gradually improved. When $m \geq 4$, the growth of ASR has become slow. Moreover, the large number of adversarial strips may lead to difficulties in the deployment to real-world scenarios. Therefore, the hyperparameter m is set to 4 during the entire optimization process.

2) Analysis of the Position, Thickness, and Color of Strip: For each adversarial strip, the optimal result is obtained by optimizing its position, thickness, and color. Hence, it is necessary to analyze the impact of these factors on attack performance. We individually optimize each factor and different combinations to observe the changes in ASR. Table III shows the ASRs of different settings on multiple detectors and datasets. The results in the first row indicate that random single-color strips without any optimization only achieve ~30% ASR in the case of the white-box attack. The attack performance is improved significantly as long as we begin to optimize either factor. Besides, the more the factors that are considered, the higher the ASR will be obtained. It is obvious that the joint optimization of position, thickness, and color outputs the best attack performance.

E. Extend to Attack Infrared Detectors

In this section, we further extend the proposed strip-based adversarial patches to attack the detectors in infrared scenes of remote sensing. It should be noted that the adversarial strip for infrared detectors is generated by optimizing its position and thickness without color. We conduct experiments on the infrared aerial detection dataset VEDAI [34] and real-world scenarios. On the VEDAI dataset, the proposed attack method achieves 97.7% and 66.5% ASRs for attacking the YOLOv5 OBB and Oriented RCNN, respectively. Fig. 9 shows the detection results of YOLOv5 OBB on adversarial examples. It can be observed that adversarial strips can successfully mislead the detector, causing the missing of the target objects. We use thermal insulation materials with low emissivity to realize the adversarial strip in the physical domain, and the experiment on real-world scenario shows the same phenomenon, as shown in Fig. 10. These results indicate that the proposed method can be effectively extended to attack infrared detectors in remote sensing.

F. Analysis of Attack Mechanism

Since the proposed single-color adversarial strip method is fundamentally distinct from the previous texture-based adversarial patches, we attempt to explore their possible attack mechanisms in remote sensing scenarios.

In this section, we design two simple experiments to explore the impacts of different adversarial patches on aerial detection models. The procedure of the first experiment is shown in Fig. 11. For a clean image containing a target object (e.g., a plane), we first edit the texture pattern of a baseball diamond to the surface of the plane. Feeding the crafted image into YOLOv5 OBB, we find that the location is predicted correctly, but the label is misclassified as a baseball diamond. We can deduce a conjecture from this result that the texture-based adversarial patch method mainly misleads the prediction of the detector's classification branch.

The procedure of the second experiment is shown in Fig. 12. For a clean image containing a target object (e.g., a plane),

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(a) (b) (c)

Fig. 12. Illustration of attacking mechanism analysis for strip-based adversarial patches. (a) Clean image. (b) Adding strip. (c) Crafted image.

we first attach one adversarial strip on the top of the plane. Feeding the crafted image into YOLOv5 OBB, we present the bounding box with the highest confidence. It can be observed that the bounding box only draws parts of the object and the position is shifted. We also can deduce a conjecture from this result that the strip-based adversarial patch method mainly misleads the prediction of the detectors' location branches.

These two conjectures may provide some inherent mechanisms for developing more efficient attack methods or enhancing the robustness of object detection models in remote sensing scenarios.

V. Conclusion

This article proposes a new physical strip attack method to deceive the detectors into missing the target objects in remote sensing. Specifically, we design a differentiable representation and an optimization method to optimize the position, thickness, and color of the adversarial strips. Comprehensive experiments in the digital domain and the physical domain demonstrate that the proposed method achieves high attack performance against white-box, black-box, and defense detectors compared with the traditional texture-based adversarial patches.

According to previous works and our analysis, the texture-based patches are indeed effective in the near distance as the rich texture information provides important features for object detection. As the increasing of distance in remote sensing scenarios, the attack performance of texture-based patches is poor when the distance extends 100 m, as presented in the above experiments. This motivates us to design a more feasible method for long-distance physical attacks. The proposed method addresses issues caused by long distances and achieves remarkable performance to attack object detection in remote sensing. Besides object detection, designing the physical attack method to study the vulnerability of other remote sensing tasks (e.g., UIU-Net [35]) is also an open and interesting direction.

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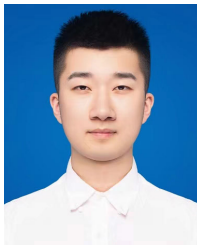
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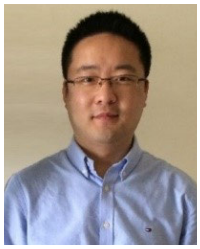
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